MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT: CKU-7/A METAL PARTS ASSEMBLY

This specification has been approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for the CKU-7/A aircraft ejection seat catapult metal parts assembly (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-413	Visual Inspection Guide for Rubber O-Rings

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD (CODE IDENT 14083)

C9254759	Container, Ammunition, Fiber, M424 for Catapult Aircraft Ejection Seat: CKU-7/A
C9254760	Box, Packing, Ammunition for Catapult, CKU-7/A in Fiber, Container, M424
C9254761	Carton, Alternative Packing, Ammunition for Catapult, Aircraft Ejection Seat CKU-7/A

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Ordnance Station, Attn: Standardization Division (611), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

F11820362	Catapult, Aircraft Ejection Seat: CKU-7/A Metal Parts Assembly
LI11820362	Inspection Equipment List for Catapult CKU-7/A
10551422	Booster Tube
10551423	Launcher Tube
10551424	Motor Tube

Copies of specifications, standards, and drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1	Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 Conformance to documents. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 First article. When specified in the contract (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

3.3 Physical properties. Physical properties specified on the drawings shall be uniform throughout each part within the tolerances. Physical properties requirements for alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The components listed in TABLE I shall withstand the corresponding internal pressure for 15 seconds (s) minimum, without application of any external load, and shall thereafter show no evidence of leakage or mechanical failure (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE I. Internal pressure requirements.

Component	Hydrostatic pressure psi, minimum
Booster tube (10551422)	11000
Launcher tube (10551423	11000
Motor tube (10551424)	5000

3.5 Shear pin strength. Shear pins shall shear when subjected to an axial load of 41 ± 5 lbs applied at the rate of 0.125 inches per minute.

3.6 Protective finish. Protective finish shall be in accordance with the applicable specification referenced on the drawings and shall apply to all surfaces regardless of configuration.

3.7 Workmanship. The requirements of workmanship shall be as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shots, inclusions, porosity or other metal defects.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part of the assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 First article inspection.

4.4.1 Initial production sample. Unless otherwise specified in the contract (see 6.2), an initial production sample of 10 sets of unassembled components and 50 shear pins from the same lot of pins used by the contractor shall be submitted to an activity designated by the procuring activity (see 6.2) for initial production examination and test. The sample shall be manufactured using the same materials, equipment, processes and procedures to be used in regular production. All parts and materials including packaging and packing shall be the same as those used for regular production and shall be obtained fromthe same source of supply. The inital production samples (exclusive of the 50 shear pins) shall be included in the quantity specified for delivery on contract if acceptable (see 6.2).

4.4.1.1 Examination and test. After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications.

4.4.1.2 Initial production sample failure. Failure of the sample to comply with requirements of the drawings and specifications shall result in initial production sample disapproval.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations of 4.6 and the tests of 4.7 performed on samples selected in accordance with 4.5.2 and 4.5.3. Sampling shall be in accordance with MIL-STD-105.

4.5.1 Lot size. For the purpose of identification and shipment, a shipping lot shall consist of 1200 metal parts assemblies, maximum, exclusive of test samples, provided the homogeneity of manufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer (see 6.2).

4.5.2 Sampling for examination. One hundred percent examination shall be performed for all critical defects. Sampling for major and minor defects shall be in accordance with TABLE II.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:56Z
Check the source to verify that this is the currentversion before use.

TABLE II. Sampling plan and individual defect criteria.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

4.5.3 Sampling for tests. Sampling for tests shall be in accordance with the following:

4.5.3.1 Hardness. Fifty units from each heat-treated lot of components shall be tested in accordance with 4.7.1 for compliance with the hardness requirements of the drawings.

4.5.3.2 Hydrostatic pressure. Each component listed in 3.4 shall be subjected to the internal hydrostatic pressure test of 4.7.2.

TABLE III. Total defect criteria.

Number of defect Classifications in Class *	Sample size			
	110	150	225	300
	Major defects allowed			
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
	Minor defects allowed			
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

*e.g. 101-119 is 19 defect classifications in a major class

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of Defects.

Classification	Defect	Method of inspection
	Firing pin (Drawing 8596909)	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	True position of shear pin hole with body	Gage
104	Symmetry of wrench holes with body	Gage
105	Diameter of striker	Gage
106	Diameter of shear pin hole, max.	Gage
107	Diameter of body, min.	Gage
108	Diameter of groove, min.	Gage
109	Length of striker	Gage
110	Radius at base of striker	Gage
111	Radial profile of striker	Gage
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of body and striker improper	Visual
115	Burr or sharp corner, except as required	Visual
116	Foreign matter	Visual
117	Diameter of Groove, max.	Gage
118	Width of Groove	Gage
119	Permissible variation in Groove Sidewalls	Gage
MINOR		
201	Diameter of wrench hole	Gage
202	Diameter of body, max.	Gage
203	Diameter of shear pin, min.	Gage
204	Depth of holes	Gage
205	Total length	Gage
206	Location of shear pin hole	Gage
207	Location of groove	Gage
208	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
209	Marking missing, incorrect or illegible	Visual
210	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
Wire Lock, Drawing D10533781		
CRITICAL		
None defined		
MAJOR		
101	Straightness	Gage
102	Width, min.	Gage
103	Thickness, min.	Gage
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
MINOR		
201	Width, max.	Gage
202	Thickness, max.	Gage
203	Thickness, at tang end	Gage
204	Height of tang	Gage
205	Height of chamfer end	Gage
206	Angle of tang	Gage
207	Radius or chamfer missing or incorrect	Visual
208	Surface finish improper	Visual
Spring, Locking, Drawing D10533782		
CRITICAL		
None defined		
MAJOR		
101	Length, solid, max.	Gage
102	Load at specified heights	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual
MINOR		
201	Diameter, outside	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Key, Drawing C10533783	
CRITICAL		
None defined		
MAJOR		
101	Height to datum	Gage
102	Radius inside	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Foreign matter	Visual
MINOR		
201	Total length	Gage
202	Width	Gage
203	Width of small angle	Gage
204	Thickness	Gage
205	Angle, small	Gage
206	Angle, large	Gage
207	Radius or chamfer missing or incorrect	Visual
208	Surface finish improper	Visual
	Lug, Drawing C10533785	
CRITICAL		
None defined		
MAJOR		
101	Perpendicularity of ends with diameter	Gage
102	Diameter	Gage
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual
MINOR		
201	Total length	Gage
202	Chamfer missing or incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Cam, Drawing F10551411	
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of thread, max.	Gage
102	Minor diameter of thread, min.	Gage
103	Concentricity of small radius with outside radius	Gage
104	Concentricity of large radius with outside radius	Gage
105	Width of groove	Gage
106	Width of flange	Gage
107	Location of groove from face	Gage
108	Location of holes	Gage
109	Angles of lead of helix	Gage
110	Metal defective	Visual
111	Protective finish inadequate or defective	Visual
112	Burr	Visual
113	Foreign matter	Visual
MINOR		
201	Outside radius	Gage
202	Length from sharp edge to groove	Gage
203	Location of groove from datum	Gage
204	Angular location of holes	Gage
205	Angular width from center line	Gage
206	Angle, primary center section	Gage
207	Location to datum	Gage
208	Radius, small, center section	Gage
209	Radius, large, center section	Gage
210	Radius or chamfer missing or incorrect	Visual
211	Marking missing, incorrect or illegible	Visual
212	Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Head, Drawing F10551412	
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of set screw thread, max.	Gage
102	Minor diameter of set screw thread, max.	Gage
103	Pitch diameter of gas intake port thread, max.	Gage
104	Minor diameter of gas intake port thread, max.	Gage
105	Concentricity of firing pin cavity with large cavity	Gage
106	Concentricity of large cavity with body	Gage
107	Concentricity of internal groove with large cavity	Gage
108	Concentricity of pitch diameter of internal thread with large cavity	Gage
109	Concentricity of external O-ring groove with body	Gage
110	Concentricity of lockwire groove with body	Gage
111	Concentricity of counterbore with gas intake thread	Gage
112	Perpendicularity of set screw alignment hole with mounting hole	Gage
113	Perpendicularity of mounting hole with body	Gage
114	Perpendicularity of gas intake port spot face to gas port	Gage
115	Perpendicularity of mounting surface with mounting hole	Gage
116	Parallelism of mounting surface	Gage
117	True position of set screw alignment hole with body	Gage
118	True position of shear pin hole with firing pin cavity	Gage
119	True position of flat bottom holes with small set screw alignment hole	Gage
120	Symmetry of small hole with lockwire groove	Gage
121	Symmetry of mounting surface with body	Gage
122	Diameter of large cavity, max.	Gage
123	Diameter of firing pin cavity, max.	Gage
124	Diameter of internal O-ring groove	Gage
125	Diameter of counterbore of gas intake port	Gage
126	Diameter of mounting hole	Gage
127	Diameter of chamfer of gas intake port	Gage
128	Diameter of thread undercut	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
129	Diameter of body	Gage
130	Diameter of external O-ring groove	Gage
131	Diameter of lock wire groove	Gage
132	Diameter of shear pin hole	Gage
133	Diameter of small set screw alignment hole	Gage
134	Pitch diameter of gas intake port thread, min	Gage
135	Depth to base of counterbore at gas intake port	Gage
136	Width of external O-ring groove, max.	Gage
137	Width of lock ring groove, max.	Gage
138	Angle of chamfer at entrance of gas inlet port	Gage
139	Pitch diameter of internal thread, max.	Gage
140	Minor diameter of internal thread, max.	Gage
141	Depth from face to internal O-ring groove	Gage
142	Depth of full thread in set screw hole	Gage
143	Metal defective	Visual
144	Protective finish inadequate or defective	Visual
145	Surface finish of chamfer on gas intake port mouth, improper	Visual
146	Burr or sharp corner	Visual
147	Foreign matter	Visual
148	Depth of effective gas port thread, min	Gage
149	Width of internal O-Ring Groove	Gage
150	Width of O-Ring Groove, min	Gage
151	Location of lock wire groove	Gage
152	Location of external O-Ring groove	Gage
153	Pitch diameter of internal thread	Gage
MINOR		
201	Pitch diameter of set screw thread, min.	Gage
202	Diameter of flat bottom holes	Gage
203	Diameter of firing pin cavity, min.	Gage
204	Diameter of large cavity, min.	Gage
205	Diameter of largest outside diameter	Gage
206	Diameter of internal counterbore at firing pin hole	Gage
207	Diameter of internal gas port hole	Gage
208	Depth of drilled cavity from end	Gage
209	Depth of firing pin cavity at firing pin hole	Gage
210	Depth of counterbore	Gage
211	Depth of large cavity	Gage
212	Depth of gas port hole	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
213	Depth of cast cavity	Gage
214	Depth of thread undercut	Gage
215	Radius on cast end	Gage
216	Length from end to face	Gage
217	Length from end to shoulder	Gage
218	Length from face to mounting end	Gage
219	Length from face to centerline mounting hole	Gage
220	Width of thread undercut	Gage
221	Width of lock wire groove, min.	Gage
222	Location of set screw hole	Gage
223	Location of shear pin set screw hole	Gage
224	Anglar location of lockwire hole	Gage
225	Location of gas port from mounting hole	Gage
226	Spot face diameter	Gage
227	Diameter of small drilled hole in groove	Gage
228	Depth of small drilled hole in groove	Gage
229	Width of mounting surface	Gage
230	Width of cutout on mounting end	Gage
231	Thickness of mounting surface	Gage
232	Thickness of cutout on mounting end	Gage
233	Diameter of cast boss	Gage
234	Diameter of large cavity	Gage
235	Length from face to backface of shoulder	Gage
236	Depth of small set screw alignment hole	Gage
237	Diameter of drilled hole at bottom of firing pin hole	Gage
238	Depth of flat bottom holes	Gage
239	Depth of large cast cavity	Gage
240	Location of flat bottom holes	Gage
241	Marking missing, incorrect or illegible	Visual
242	Radius or chamfer missing or incorrect	Visual
243	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Mount, Drawing D10551413	
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of thread, max.	Gage
102	Minor diameter of thread, max.	Gage
103	Concentricity of pitch diameter of thread with counterbore	Gage
104	Concentricity of counterbore with small cavity	Gage
105	Concentricity of datum with thread counterbore	Gage
106	Concentricity of mounting lugs	Gage
107	True position of mounting lug with datum	Gage
108	Symmetry of min length of mounting lugs with datum	Gage
109	Symmetry of max length of mounting lugs with datum	Gage
110	Perpendicularity of face with counterbore	Gage
111	Diameter of small cavity, max.	Gage
112	Diameter of counterbore, max.	Gage
113	Diameter of groove	Gage
114	Diameter of shoulder	Gage
115	Diameter of mounting lugs	Gage
116	Width of groove, max.	Visual
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual
119	Surface finish of 63 or better, improper	Visual
120	Burr or sharp corner	Visual
121	Foreign matter	Visual
122	Width of groove, min.	Gage
123	Location of groove	Gage
MINOR		
201	Pitch diameter of thread, min.	Gage
202	Diameter of small cavity, min.	Gage
203	Diameter of counterbore, min.	Gage
204	Diameter of body, max. (large area)	Gage
205	Diameter of small hole	Gage
206	Diameter of thread undercut	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
207	Depth of small cavity	Gage
208	Depth of counterbore	Gage
209	Length across mounting holes	Gage
210	Total length	Gage
211	Length from small end to face	Gage
212	Length from center to flat	Gage
213	Width between mounting lug flats	Gage
214	Width of undercut	Gage
215	Width of flat	Gage
216	Width at bottom	Gage
217	Location of datum diameter of taper from face	Gage
218	Location of mounting lugs from face	Gage
219	Radius, outside	Gage
220	Location of hole	Gage
221	Radius or chamfer missing or incorrect	Visual
222	Marking missing, incorrect or illegible	Visual
223	Surface finish improper except as otherwise classified	Visual
	Nozzle, Drawing DI0551414	
CRITICAL		
1	Thickness of wall at thin section (A-A)	Gage
MAJOR		
101	Concentricity of pivot hole with other pivot hole	Gage
102	Concentricity of spherical surface with datum diameter	Gage
103	Concentricity of small datum with spherical surface	Gage
104	Concentricity of small bore with datum diameter	Gage
105	True position pivot hole with sperical diameter	Gage
106	Symmetry of pivot holes with spherical diameter	Gage
107	Diameter of spherical surface	Gage
108	Diameter at datum on large tampered bore	Gage
109	Diameter at datum on small chamfer	Gage
110	Diameter of small bore	Gage
111	Diameter of pivot holes	Gage
112	Angle of cam surface, large	Gage
113	Angle of cam surface, small	Gage
114	Location of small datum from face	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
115	Location of large datum from face	Gage
116	Metal defective	Visual
117	Protective finish inadequate or defective	Visual
118	Surface finish or surfaces calling for 63 finish or better	Visual
119	Burr or sharp corner	Visual
120	Foreign matter	Visual
121	Angle to wall of cutout, max. (68 ± 1°)	Gage
122	Location of cavity	Gage
MINOR		
201	Diameter of small hole	Gage
202	Diameter of large groove	Gage
203	Depth of slot	Gage
204	Total length	Gage
205	Width of flange from center	Gage
206	Thickness of flange at slot	Gage
207	Width of small bore, min.	Gage
208	Width of slot	Gage
209	Width of cavity	Gage
210	Width between bottom of pivot holes	Gage
211	Width of groove	Gage
212	Width of surface adjacent to cam area	Gage
213	Location of pivot hole from face of flange	Gage
214	Location of pivot hole from inside face of flange	Gage
215	Location from pivot hole to far edge of cavity	Gage
216	Location of small hole	Gage
217	Angle of chamfer at small end	Gage
218	Angle of lead for cam surface	Gage
219	Radius, outside edge	Gage
220	Radius of bottom flange	Gage
221	Radius of cam surface	Gage
222	Radius of inside wall, thin section	Gage
223	Large radius	Gage
224	Length of angle from end of cam (C-C)	Gage
225	Centerline from end of cam (C-C)	Gage
226	Marking missing, incorrect or illegible	Visual
227	Radius or chamfer missing or incorrect	Visual
228	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
Pin, Shear, Drawing C10551415		
CRITICAL		
None defined		
MAJOR		
101	Diameter, min.	Gage
102	Total length, min.	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
MINOR		
201	Diameter, max.	Gage
202	Total length, max.	Gage
203	Surface finish improper	Visual
204	Radius or chamfer missing or incorrect	Visual
Release, Drawing D10551416		
CRITICAL		
None defined		
MAJOR		
101	Concentricity of flange with end	Gage
102	Concentricity of small diameter with end	Gage
103	Concentricity of flange end diameter on right end of view with center body diameter	Gage
104	Concentricity of groove with end diameter	Gage
105	Diameter of flange on left end of view	Gage
106	Diameter of body at center	Gage
107	Diameter of body at groove end	Gage
108	Diameter of groove	Gage
109	Thickness of wall, min.	Gage
110	Width of groove	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of 63 or better, improper	Visual
114	Burr or sharp edge	Visual
115	Foreign matter	Visual
116	Location of flange	Gage
117	Width of O-Ring groove	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
MINOR		
201	Diameter of hole	Gage
202	Diameter at thin section	Gage
203	Diameter at thick section	Gage
204	Depth of hole	Gage
205	Length of chamfer	Gage
206	Total length	Gage
207	Width of flange	Gage
208	Angle of chamfer	Gage
209	Location of groove from end	Gage
210	Marking missing, incorrect or illegible	Visual
211	Radius or chamfer missing or incorrect	Visual
212	Surface finish improper, except as otherwise classified	Visual
	Retainer, Drawing F10551418	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of large radius with outside diameter	Gage
102	Concentricity of small radius with outside diameter	Gage
103	Concentricity of large cavity with outside diameter	Gage
104	Concentricity of 2 outside diameters with each other	Gage
105	Concentricity of small inside diameter with large cavity	Gage
106	Concentricity of small groove with large cavity	Gage
107	Concentricity of large groove with large cavity	Gage
108	Concentricity of outside edge of spherical cavity to spherical cavity	Gage
109	Concentricity of lock wire groove with outside diameter	Gage
110	Concentricity of external O-ring groove with outside diameter	Gage
111	True position of spherical diameter with pivot hole	Gage
112	True position of pivot hole with outside diameter	Gage
113	True position of set screw hole with outside diameter	Gage
114	True position of large set screw hole with small set screw hole (A-A)	Gage
115	True position of pivot hole with outside diameter (View B)	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
116	Perpendicularity of small set screw hole with pivot hole	Gage
117	Diameter of body	Gage
118	Diameter of external O-ring groove	Gage
119	Diameter of lock wire groove	Gage
120	Diameter, spherical	Gage
121	Diameter of O-ring groove spherical surface (View B)	Gage
122	Diameter of internal O-ring groove	Gage
123	Diameter of large bore	Gage
124	Diameter of internal retaining ring groove	Gage
125	Diameter of pivot holes (View B)	Gage
126	Radius on cam surface (small)	Gage
127	Radius on cam surface (large)	Gage
128	Radius of U-shaped hole	Gage
129	Symmetry of lockwire hole with groove	Gage
130	Location of datum in groove (spherical surface) from pivot hole (View B)	Gage
131	Angle from vertical centerline of pivot line	Gage
132	Metal defective	Visual
133	Protective finish inadequate or defective	Visual
134	Surface finish of 63 or better improper	Visual
135	Burr or sharp corner	Visual
136	Foreign matter	Visual
137	Location of O-Ring groove	Gage
138	Width of O-Ring groove in spherical surface (View B)	Gage
139	Width of internal O-Ring groove	Gage
140	Width of external O-Ring groove	Gage
141	Width of cam groove (View B)	Gage
142	Width of cam shoulder (track) (View B)	Gage
143	Width to beginning of external recess	Gage
144	Width to end of external recess	Gage
145	Location of internal O-Ring groove	Gage
146	Location of pivot holes from face of shoulder (View B)	Gage
147	Location of external O-Ring groove	Gage
148	Location of groove in spherical surface from pivot hole (View B)	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
MINOR		
201	Diameter of large internal cavity	Gage
202	Diameter of small bore	Gage
203	Diameter of large undercut on O.D.	Gage
204	Diameter of small set screw hole	Gage
205	Diameter of large set screw hole	Gage
206	Diameter of lock wire tang hole	Gage
207	Diameter of small bleed hole	Gage
208	Depth of large cavity	Gage
209	Depth of small set screw hole	Gage
210	Depth of large set screw hole (A-A)	Gage
211	Depth of lock wire tang hole	Gage
212	Width of retaining ring groove	Gage
213	Width of lock wire groove	Gage
214	Width of cutout	Gage
215	Location of small bleed hole	Gage
216	Location of retaining ring groove	Gage
217	Location of radius U-shaped hole	Gage
218	Location of shoulder (View B)	Gage
219	Location of lock wire groove	Gage
220	Location of set screw holes	Gage
221	Angular location of lock wire tang hole	Gage
222	Angular location of large set screw holes	Gage
223	Angular location of screw holes (A-A)	Gage
224	Angle at end	Gage
225	Radius of cam groove (View B)	Gage
226	Width of large internal cavity	Gage
227	Depth of large bore from face	Gage
228	Total length	Gage
229	Radius of chamfer missing or incorrect, except as otherwise classified	Visual
230	Marking missing, incorrect or illegible	Visual
231	Surface finish improper, except as otherwise classified	Visual
	Sleeve, Drawing D10551420	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of large outside diameter with small outside diameter	Gage
102	Concentricity of angular surface with small outside diameter	Gage
103	Concentricity of groove with small outside diameter	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
104	Diameter of groove	Gage
105	Diameter of body	Gage
106	Outside diameter, small body	Gage
107	Width of groove, max.	Gage
108	Width of end from center	Gage
109	Angle of bore	Gage
110	Metal defective	Visual
111	Protective finish inadequate or defective	Visual
112	Surface finish of groove, body and flanges	Visual
113	Burr or sharp corner	Visual
114	Foreign matter	Visual
MINOR		
201	Diameter of bore	Gage
202	Length from end to shoulder	Gage
203	Length from top to shoulder	Gage
204	Height at small end	Gage
205	Height of tang	Gage
206	Width of shoulder	Gage
207	Width of tang	Gage
208	Location of groove	Gage
209	Location of tang	Gage
210	Angle of tang	Gage
211	Radius at end	Gage
212	Angle of short end of bore	Gage
213	Thickness of tang	Visual
214	Marking missing, incorrect or illegible	Visual
215	Radius or chamfer missing or incorrect	Visual
216	Surface finish improper except as otherwise classified	Visual
	Track, Drawing D10551421	
CRITICAL		
None defined		
MAJOR		
101	Roundness of inside diameter	Gage
102	Concentricity of thin outside diameter with inside diameter	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:56Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
103	Concentricity of extreme outisde diameter with inside diameter	Gage
104	Concentricity intermediate I.D. with small I.D. (C-C)	Gage
105	Concentricity of large I.D. with small I.D. (B-B)	Gage
106	Perpendicularity base with small inside diameter	Gage
107	Inside diameter (A-A)	Gage
108	Right hand spiral (bottom view)	Gage
109	Left hand spiral	Gage
110	Location of groove (B-B)	Gage
111	Width of groove (B-B)	Gage
112	Location of centerline of groove from end (bottom view)	Gage
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Surface finish of 63 RMS or better improper	Visual
116	Burr or sharp corner	Visual
117	Foreign matter	Visual
118	Surface around thru holes conform to ID contour	Visual
MINOR		
201	Diameter of outside thin section (A-A)	Gage
202	Diameter of small through holes	Gage
203	Diameter outside of casting (left end view)	Gage
204	Diameter inside (C-C)	Gage
205	Diameter outside (C-C)	Gage
206	Diameter inside (B-B)	Gage
207	Diameter outside (top view)	Gage
208	Diameter of small countersinks	Gage
209	Depth of groove	Gage
210	Length of circular section (bottom view)	Gage
211	Length to edge of large radius	Gage
212	Length to edge of 78° angle	Gage
213	Length from edge of flat to end of left side (top view)	Gage
214	Length of milled surface	Gage
215	Length from run out of milled surface to end (top view)	Gage
216	Total length	Gage
217	Length to largest inside diameter	Gage
218	Length to beginning of groove (top view)	Gage

Source: https://assist.dla.mil -- Downloaded: 2025-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defect. - Continued

Classification	Defect	Method of inspection
219	Length to end of groove (top view)	Gage
220	Width of flat bottom right side (bottom view)	Gage
221	Width of edge (D-D)	Gage
222	Width of edge (C-C)	Gage
223	Width of thick section each side (B-B)	Gage
224	Location of edge of radius (bottom view)	Gage
225	Location thickness each side (B-B)	Gage
226	Location of holes from end (top view)	Gage
227	Location of groove from outside diameter (right end view)	Gage
228	Angular location thin section (top of left end view)	Gage
229	Angular location thin section bottom (2 places) (left end view)	Gage
230	Angular location shoulder (right end view)	Gage
231	Angular location to flat at end of groove (right end view)	Gage
232	Angle of tapered edge (C-C)	Gage
233	Angle from centerline of casting to edge left (C-C)	Gage
234	Angle from center line to edge of right side (C-C)	Gage
235	Angle at bottom (bottom view)	Gage
236	Angle at bottom edge	Gage
237	Angle at end right side (bottom view)	Gage
238	Angle (D-D)	Gage
239	Length (top view)	Gage
240	Angular location of holes to centerline (left end view)	Gage
241	Large radius on end (bottom view)	Gage
242	Marking missing, incorrect or illegible	Visual
243	Radius or chamfer missing or incorrect	Visual
244	Surface finish improper except as otherwise classified	Visual
Tube, Booster Assembly, Drawing D10551422		
CRITICAL None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
MAJOR		
101	Pitch diameter of thread, min.	Gage
102	Major diameter of thread, min.	Gage
103	Concentricity of pitch diameter of thread with outside diameter	Gage
104	Concentricity of outside diameter with inside diameter	Gage
105	Concentricity of small outside diameter (groove end) with large outside diameter	Gage
106	Concentricity of groove with outside diameter	Gage
107	Concentricity midpoint outside diameter with end	Gage
108	Diameter, outside diameter (groove end)	Gage
109	Diameter outside diameter thread end (thread pilot)	Gage
110	Diameter of large bore for cartridge, max.	Gage
111	Diameter of groove, min.	Gage
112	Diameter of outside of center, min.	Gage
113	Thickness of wall, min. (groove end)	Gage
114	Thickness of wall (threaded end), min.	Gage
115	Depth of pellet holes	Gage
116	Diameter of pellet holes	Gage
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual
119	Surface finish of groove and thread pilot improper	Visual
120	Burr or sharp corner	Visual
121	Foreign matter	Visual
122	Width of groove	Gage
123	Protrusion of pellet over thread, max	Gage
124	Pellet missing, loose or improperly inserted	Manual/ Visual
MINOR		
201	Pitch diameter of thread, max.	Gage
202	Major diameter of thread, max.	Gage
203	Diameter of large bore, min.	Gage
204	Diameter of small bore	Gage
205	Diameter of groove, max.	Gage
206	Depth of large bore	Gage
207	Total length	Gage
208	Length of thread	Gage
209	Length from face to small OD	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
210	Location of thread	Gage
211	Location of thread	Gage
212	Location of pellet holes	Gage
213	Blend at bottom of cartridge bore improper	Visual
214	Radius or chamfer missing or incorrect	Visual
215	Marking missing, incorrect or illegible	Visual
216	Surface finish improper except as otherwise classified	Visual
217	Location of pellets	Gage
218	Angular location of pellets	Gage
Tube, Launcher, Assembly, Drawing 10551423		
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of thread, min.	Gage
102	Major diameter of thread, min.	Gage
103	Concentricity of pitch diameter of thread with thread pilot	Gage
104	Concentricity of thread pilot with inside diameter	Gage
105	Symmetry of slot with inside diameter	Gage
106	Straightness of tube	Gage
107	Diameter, outside, threaded end (thread pilot)	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
108	Diameter, inside, threaded end	Gage
109	Thickness of wall, min.	Gage
110	Depth of pellet holes	Gage
111	Diameter of pellet holes	Gage
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish inside of tube, threaded end	Visual
115	Foreign matter	Visual
116	Protrusion of pellet over thread, max.	Gage
117	Pellet missing, loose or improperly inserted	Manual/ Visual
MINOR		
201	Pitch diameter of thread, max.	Gage
202	Diameter, outside at end	Gage
203	Diameter, inside at end	Gage
204	Total length	Gage
205	Length of thread pilot	Gage
206	Length of thread	Gage
207	Width of slot	Gage
208	Location of slot	Gage
209	Length of slots	Gage
210	Location of pellet holes	Gage
211	Marking missing, incorrect or illegible	Visual
212	Radius of chamfer missing or incorrect	Visual
213	Surface finish improper, except as otherwise classified	Visual
214	Location of pellet	Gage
215	Angular location of pellet	Gage
	Tube, Motor, Drawing F10551424	
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of small set screw thread, max.	Gage
102	Minor diameter of small set screw thread, max.	Gage
103	Pitch diameter of large set screw thread, max.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
104	Minor diameter of large set screw thread, max.	Gage
105	Concentricity of inside diameter at each end with opposite end	Gage
106	Concentricity of outside diameter with inside diameter	Gage
107	Concentricity of internal grooves with inside diameter	Gage
108	True position of set screw thread with I.D.	Gage
109	True position of thread with thread and I.D.	Gage
110	True position of large set screw thread with small set screw thread	Gage
111	Symmetry of width of slots with grooves	Gage
112	Diameter, inside at center	Gage
113	Diameter, inside at ends	Gage
114	Diameter of internal groove	Gage
115	Width of internal groove, max.	Gage
116	Width of slot, max.	Gage
117	Thickness of wall, min.	Gage
118	Diameter, outside	Gage
119	Metal defective	Visual
120	Protective finish inadequate or defective	Visual
121	Surface finish on inside diameter improper	Visual
122	Burr	Visual
123	Foreign matter	Visual
MINOR		
201	Pitch diameter of small set screw thread, min.	Gage
202	Pitch diameter of large set screw thread, min.	Gage
203	Total length	Gage
204	Length of slot	Gage
205	Length of blended slot	Gage
206	Length of finish on inside, diameter	Gage
207	Width of internal groove, min.	Gage
208	Width of slot, min.	Gage
209	Location of set screw holes	Gage
210	Location of groove	Gage
211	Location of slot	Gage
212	Marking missing, incorrect, or illegible	Visual
213	Radius or chamfer missing or incorrect	Visual
214	Surface finish improper except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Plate, Drawing C10551433	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of hole with body	Gage
102	Diameter of body, min.	Gage
103	Diameter of hole, min.	Gage
104	Thickness	Gage
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr or sharp corner	Visual
108	Foreign matter	Visual
MINOR		
201	Diameter of body, max.	Gage
202	Diameter of hole, max.	Gage
203	Marking missing, incorrect or illegible	Visual
204	Radius missing or incorrect	Visual
205	Surface finish improper	Visual
	Spring, Grain, Drawing 11731754	
CRITICAL		
None defined		
MAJOR		
101	Parrallelism of ends	Gage
102	Loat at specified heights	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual
MINOR		
201	Diameter, inside, large end	Gage
202	Diameter inside, small end	Gage
203	Radius missing or incorrect	Visual
204	Direction of turn of helix	Visual
205	Surface finish improper	Visual
206	Interference between coils at closed heights	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
Catapult, Aircraft Ejection Seat, CKU-7/A, Metal Parts Assembly Drawing 11820362		
CRITICAL		
None defined		
MAJOR		
101	Any part missing, except as otherwise classified	Visual
102	Improper assembly	Visual
103	Marking missing, incorrect or illegible	Visual
MINOR		
201	Fungus or moisture resistant varnish missing	Visual
202	Shipping plug missing	Visual
203	Protective finish inadequate or defective	Visual
Container, Ammunition, Fiber, Drawing C9254759		
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201	Cap between body and cover, max.	Manual
202	Contents loose	Manual
203	Metal ends loose or distorted	Manual
204	Cut or damaged through impregnated layers	Visual
205	Marking missing, incorrect or illegible	Visual
206	Tape incomplete or badly wrinkled	Visual
207	Length of tear tab, min.	Visual
208	Color of tape, incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects. - Continued

Classification	Defect	Method of inspection
Box, Packing, Ammunition Level A Drawing C9254760		
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201	Contents loose	Manual
202	Hardware missing, loose, broken or improperly engaged	Manual/ Visual
203	Strapping missing, loose, broken or improperly assembled	Manual Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect or illegible	Visual
207	Number of units incorrect	Visual
208	Cap seal missing, unsealed or improperly positioned	Visual
Box, Packing Alternative, Level B Drawing C9254761		
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect or illegible	Visual
205	Tape incomplete	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.5.3.3 Shear pin strength. Fifty pins from each lot of shear pins usid in the contract shall be tested for shear strength in accordance with 4.7.3.

4.5.3.4 Protective finish. The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable documents listed on the drawings (see 4.7.4).

4.5.3.5 Inspection equipment. The examination and tests shall be made using equipment listed on LI 11820362.

4.6 Quality conformance examination. The quality conformance examination shall be performed on a defect basis in accordance with the classification of defects contained in TABLE IV using the sampling plan and associated criteria in TABLES II and III. For an individual defect (single characteristic), the criteria in TABLE II shall be used. For total defects within a major or minor class the criteria of TABLE III shall be used. All nonconforming material shall be removed from the lot. Inpsection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ANSI B46.1, shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification shall be used in performing visual examination for such defects as burrs and foreign matter.

4.7 Test methods.

4.7.1 Hardness. Prior to the application of the protective finish, a hardness testing machine shall be used to verify compliance with drawing requirements. Failure of any unit to comply with any drawing requirement shall be cause for rejection of the lot.

4.7.2 Hydrostatic pressure. Prior to the application of the protective finish, each component listed in 3.4 shall be individually mounted in a test device, using the fixture for sealing all openings and subjected to the applicable internal hydrostatic pressure specified in TABLE I to determine compliance with 3.4. The test device used shall internally seal the components within 1.5 inches of each end without applying an axial load other than that applied by the liquid pressure. Any evidence of leakage or mechanical failure shall be cause for rejection of that component.

4.7.3 Shear pin strength. The shear pin strength test shall be performed after application of the protective finish. The pin shall be inserted in the fixture and tested in conjunction with a tensile testing machine. The axial load shall be applied at the rate of 0.125 inches per minute. Failure of any pin to shear within the range of 36 to 46 lbs. shall be cause for rejection of the lot of pins.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.7.4 Protective finish. Test procedures and acceptance criteria for protective finish shall be in accordance with the applicable documents referenced on the drawings.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be Level A or Level B as specified by the procuring agency (see 6.2).

5.1.1 Level A. Preservation, packaging, packing and marking shall be in accordance with Drawings C9254759 and C9254760.

5.1.2 Level B. Preservation, packaging, packing and marking shall be in accordance with Drawings C9254759 and C9254761.

6. NOTES

6.1 Intended use. The CKU-7/A, aircraft ejection seat catapult metal parts assembly, described herein, is intended to be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems used in propeller driven aircraft.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements:

a. Title, number and date of this specification.
b. Whether a first article sample is required (see 3.2 and 6.4).
c. Quantity required (including first article samples).
d. Assigned activity for first article inspection (see 4.4).
e. The level of packaging required (see 5.1).

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD Form 1664) which will be documented in the Applicable ADL (Authorized Data List).

6.3 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurence of which is other than a design function.

6.4 First article sample waiver. First article samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2.1(b)), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract or order.

Custodian:
NAVY - OS

Preparing activity:
NAVY - OS

Project No. 1377-N639

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.